Name _____ Period _____ Date _____

4-3 Lesson Quiz: Multiplying and Dividing Rational Expressions short form

Question numbers match the Savvas Lesson Quiz. Questions 1 and 3 are not on this form. Show your factoring for every question.

2. Simplify $\dfrac{(x+5)(x^2-25)}{(x+5)^2(x-5)^2}$. What is the domain?

Ⓐ $\dfrac{x+5}{x-5}$; all real numbers except -5 and 5

Ⓑ $\dfrac{1}{x-5}$; all real numbers except -5 and 5

Ⓒ $\dfrac{1}{x+5}$; all real numbers except 0

Ⓓ $\dfrac{x-5}{x+5}$; all real numbers except -5

Work: factor, then name the values that make the original denominator zero.

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4. What is the quotient of $\dfrac{9-x^2}{3x}$ and $\dfrac{x^2+6x+9}{3x}$?

Ⓐ $\dfrac{x-3}{3-x}$ Ⓑ $\dfrac{x-3}{x+3}$

Ⓒ $\dfrac{3-x}{x+3}$ Ⓓ $\dfrac{x+3}{x-3}$

Work: write the division as a multiplication by the reciprocal, then factor.

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5. The area of a rectangle is $\dfrac{x^2-4}{2x}$ in.^{2} and its length is $\dfrac{(x+2)^2}{2}$ in. What is the width in inches?

Ⓐ $\dfrac{x(x+2)}{x-2}$ Ⓑ $\dfrac{x+2}{x(x+2)}$

Ⓒ $\dfrac{x-2}{x(x-2)}$ Ⓓ $\dfrac{x-2}{x(x+2)}$

Work: width = area ÷ length.

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Scoring: Question 2 and Question 4 are 10 points each. Question 5 is a bonus of up to 5 points. Total out of 20.

4-3 Lesson Quiz, short form: answer key

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2. B, $\dfrac{1}{x-5}$; all real numbers except -5 and 5. $x^2 - 25 = (x+5)(x-5)$, so the expression is $\dfrac{(x+5)(x+5)(x-5)}{(x+5)^2(x-5)^2} = \dfrac{1}{x-5}$. The domain comes from the original denominator $(x+5)^2(x-5)^2$: $x \neq -5$ and $x \neq 5$. Choices A, C and D are not equivalent to the expression; C and D also state the wrong domain (C leaves out only 0, D leaves 5 in). Savvas quiz item 2, DOK 2, skill A85; taught by Example 2. **10 points:** 6 for B, 4 for the work (both factorizations shown and the excluded values named from the original denominator).

4. C, $\dfrac{3-x}{x+3}$. $\dfrac{9-x^2}{3x} \cdot \dfrac{3x}{x^2+6x+9} = \dfrac{(3-x)(3+x)\cdot 3x}{3x\,(x+3)(x+3)} = \dfrac{3-x}{x+3}$ (domain, not asked: $x \neq 0$, -3). Choice B is the sign slip $x-3$ for $3-x$ (they are opposites); D is the reciprocal of B. Savvas quiz item 4, DOK 2, skill A85; taught by Example 5. **10 points:** 6 for C, 4 for the work (the reciprocal written, $9-x^2$ and x^2+6x+9 factored).

5. D, $\dfrac{x-2}{x(x+2)}$. Width = area ÷ length = $\dfrac{x^2-4}{2x} \cdot \dfrac{2}{(x+2)^2} = \dfrac{(x+2)(x-2)\cdot 2}{2x\,(x+2)(x+2)} = \dfrac{x-2}{x(x+2)}$. Savvas quiz item 5, DOK 2, skill A85; taught by Example 5 (division), rehearsed by the packet's Practice (TE Practice 37, base of a parallelogram = area ÷ height). **Bonus 5:** 5 for D with the division set up as area times the reciprocal of the length; 2 for D with no work.

Items left off this form (available for a retake or Form B): Savvas item 1, simplify $\dfrac{x^2-2x}{(x-2)(x+3)}$ (answer C, $\dfrac{x}{x+3}$; DOK 2, A85; Example 2); Savvas item 3, simplify $\dfrac{ab^2-a^2b}{ab} \cdot \dfrac{a^2b^2}{a-b}$ (answer A, $-a^2b^2$; $a \neq 0$, $b \neq 0$; DOK 2, A85; Example 3, cut in the fast packet). The TE's choice A omits $a \neq b$, which the original denominator $a-b$ requires; accept A and mention it.

IXL replacement: SmartScore 70 or above on 37N, 79Q and MG2 replaces this quiz with 20/20.